



**FUZE™**

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BRAND & COMPLIANCE  
**STANDARDS**

# Strategic Brand Objectives

Protecting the FUZE® brand — not only in how it looks, but in how it is messaged, claimed, and understood — is essential to everything we build together. Our goal is a single, consistent brand voice and story across every partner, so that the FUZE® name reliably signals authority, performance, and trust to both our customers and the consumers they serve.

FUZE® Technologies maintains a Brand & Compliance Program that reinforces our brand agreements and guidelines to strengthen brand equity, create demand, communicate a clear point of difference, and ensure quality, long-lasting partnerships. This program tells your customers that you are using a best-in-class, independently validated, and genuinely sustainable antimicrobial technology — one engineered to elevate the performance of the textiles you make.

Since our first edition in 2021, FUZE® has matured into a federally **EPA-registered** platform with **California EPA** approval, a four-tier product system, and a body of independent third-party validation. This edition brings our standards current with that reality — most importantly, it codifies the **claim discipline** that keeps every partner's marketing both compelling and compliant under U.S. EPA rules.

Please treat these guidelines as binding whenever you use the FUZE® name, marks, or approved messaging. Together, we are building a cleaner, healthier, more sustainable world — one textile at a time.

## WHAT THIS EDITION ADDS

- EPA Treated Article claim framework & approved claim lists
- Current certifications: EPA (federal), California EPA, PFAS-free
- Brand language standard — the FUZE® metamaterial vocabulary
- F1–F4 product system & application methods
- Global market claims — US · EU · Canada · China
- ASTM E2149 vs. AATCC 100 testing position
- Refreshed identity, color, and digital standards

## TEAM FUZE™

FUZE® Technologies · Salt Lake City, Utah



Organized in four parts: **Compliance** (what may be said), **Identity** (how the brand looks and sounds), **Product & Partnership**, and **Global Compliance** (market-by-market claim rules).

## PART ONE — COMPLIANCE

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FUZE® is a non-leaching antimicrobial textile treatment that bonds a proprietary **metamaterial** permanently into fiber surfaces during standard finishing.

Unlike surface coatings or leaching biocides, FUZE® does not wash out, does not migrate to skin, and requires no PFAS, binders, or chemical curing. The bond is **permanent at every application level** — it works through direct contact, so protection holds through commercial laundering instead of fading like leaching treatments.

## CERTIFICATIONS & STANDARDS

EPA Registered · Federal

California EPA Approved

OEKO-TEX® Standard 100 Class I

bluesign® system partner

ZDHC conformant

PFAS-Free

AATCC 100

ASTM E2149

AATCC 30

ISO 20743

ISO 18184

## THE FOUR-TIER SYSTEM — PERMANENT AT EVERY LEVEL

| F1               | F2          | F3         | F4          |
|------------------|-------------|------------|-------------|
| <b>1.0</b>       | <b>0.75</b> | <b>0.5</b> | <b>0.25</b> |
| MG/KG · FLAGSHIP | MG/KG       | MG/KG      | MG/KG       |
| Full Spectrum    | Advanced    | Core       | Foundation  |

**The bond is permanent at every tier.** Application level sets antimicrobial test performance, the ancillary benefit stack (reduced water surface tension, faster drying, UV), and price — not durability. F1 delivers the full stack; lower tiers are better-priced, not lower quality.

### MADE RESPONSIBLY

Manufactured at a solar-capable facility in Salt Lake City, Utah from recycled-electronics feedstock, then delivered as a water-based finishing liquid applied on standard equipment — exhaust, pad-dry-cure, or spray.

### ONE PLATFORM, ONE VOICE

Applied across hospitality, healthcare, activewear, performance apparel, and home textiles. Every partner uses the same marks, the same vocabulary, and the same approved claims.



## PART ONE · COMPLIANCE

## Certifications & the Treated Article Framework

FUZE® is regulated in the United States as an antimicrobial pesticide. Textiles treated with FUZE® are marketed under the U.S. EPA **Treated Article Exemption** (FIFRA / EPA PR Notice 2000-1), which governs what every partner may — and may not — claim.

## CURRENT CERTIFICATIONS &amp; REGISTRATIONS

| CERTIFICATION            | SCOPE                                    | WHAT IT MEANS FOR PARTNERS                                                                            |
|--------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------|
| EPA Registered — Federal | U.S. EPA antimicrobial registration      | Federally registered antimicrobial. Registration details shared with partners under agreement.        |
| California EPA Approved  | CA Dept. of Pesticide Regulation         | Cleared for sale in California — a differentiator for activewear, infant, and healthcare positioning. |
| OEKO-TEX® Standard 100   | Class I — direct skin contact            | Validated for all textile categories, including baby and infant products.                             |
| bluesign® system partner | Approved product · on-site factory audit | Facility passed an on-site bluesign® audit; recognized for responsible chemistry management.          |
| ZDHC conformant          | MRSL — self-declared                     | Self-declared conformance to the ZDHC MRSL; formal verification in progress.                          |
| PFAS-Free                | Composition                              | No per- or polyfluoroalkyl substances — aligned with tightening U.S./EU restrictions.                 |

**THE TREATED ARTICLE EXEMPTION — IN ONE PARAGRAPH**

Because FUZE® is registered as the *pesticide*, a textile *treated with* FUZE® may claim only to **protect the article itself** — resisting odor-causing bacteria, mildew, and deterioration. It may **not** make public-health claims (wearer disease protection, “kills 99.9%,” pathogen kill rates) without its own product-specific registration. Overstating scope creates FIFRA Section 12 exposure.

**Retired in this edition.** The 2021 reference to FDA EUA *EUA200344* is obsolete and must no longer be used — COVID-era EUAs have lapsed. FUZE® compliance now rests on EPA registration and the Treated Article framework above.



Use the **green** language freely on consumer-facing materials. Treat **amber** language as technical — permitted only in clearly-labeled validation contexts, never as a consumer headline. **Red** language is prohibited on treated-article marketing without a product-specific public-health registration. When in doubt, request approval at [brand@fuze47.com](mailto:brand@fuze47.com).

| ● Approved · Green                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ● Technical · Amber                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | ● Prohibited · Red                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Inhibits the growth of odor-causing bacteria</li> <li>Inhibits the growth of mildew &amp; mold that cause product deterioration</li> <li>Keeps fabric fresher, longer</li> <li>Durable, long-lasting freshness protection</li> <li>Won't wash out — bonds permanently to the fiber</li> <li>Non-leaching</li> <li>PFAS-free</li> <li>OEKO-TEX® Class I · safe for skin contact</li> <li>Factual certification statements (EPA registered, bluesign®)</li> </ul> | <ul style="list-style-type: none"> <li>Antifungal / anti-mildew efficacy (tie to AATCC 30 / ASTM G21)</li> <li>Antiviral activity (ISO 18184 data only — never a disease-protection claim)</li> <li>Durability — “permanently bonded; efficacy retained through repeated commercial laundering” (third-party tested)</li> <li>Log-reduction / efficacy figures in <i>labeled</i> technical reports</li> </ul> <p>Amber claims live in spec sheets, lab reports, and B2B technical decks — supporting the green claims, never replacing them as consumer headlines.</p> | <ul style="list-style-type: none"> <li>“Antibacterial / bactericidal / germicidal” as a product claim</li> <li>“Kills 99.9%” or any kill-rate percentage</li> <li>Named-pathogen claims (MRSA, Staph, E. coli)</li> <li>Protects the wearer from illness or infection</li> <li>Any virus / COVID-19 protection claim</li> <li>“EPA-approved” / “EPA-validated” durability or efficacy</li> <li>silver, nano, nanoparticle, silver-ion, Ag</li> </ul> |

**Why “kills 99.9%” is off-limits.** Specific kill claims convert a treated article into a “public-health pesticide” in the EPA’s eyes, requiring a separate product registration. The same lever we use against competitors who overclaim applies to us.

**Why never “EPA-validated.”** The EPA registers chemistry — it does not certify durability or efficacy. FUZE®’s permanent bond and antimicrobial performance are *third-party* tested (AATCC 100 / ASTM E2149 / ISO 20743), with reports on request. Say “*third-party validated*,” not “EPA-validated.”



## PART TWO · BRAND IDENTITY

## How We Talk About FUZE

The FUZE® brand is built on a precise vocabulary. In all marketing, web, and customer-facing content, our active technology is the **metamaterial** — never “silver,” “nano,” or “ion.” This is both a positioning choice and a compliance discipline. The distinction that matters is our *physical form and bonding mechanism*, not the underlying element.

## THE VOCABULARY STANDARD

| NEVER SAY               | ALWAYS SAY             |
|-------------------------|------------------------|
| silver                  | FUZE®                  |
| nanoparticle / nano     | metamaterial           |
| silver-ion / silver ion | non-leaching treatment |
| nano-silver, Ag         | FUZE® metamaterial     |
| “water-based silver”    | high-density allotrope |

**One exception:** regulatory & technical documents (SDS, CIL, ARSL) may name the chemistry by CAS number. Marketing, web, and partner collateral may not.

## NAMING THE COMPANY

**FUZE® Technologies** — the umbrella name for cross-vertical positioning (textiles, thermal, IR-deflection). Use for press, partnerships, and general brand.

**FUZE® Biotech** — use where the context is strictly microbiology, chemistry, or regulatory filings.

**FUZE®** — the product itself. Tiers are written **F1**, **F2**, **F3**, **F4** (e.g. “FUZE® F1 Full Spectrum”).

## TONE OF VOICE

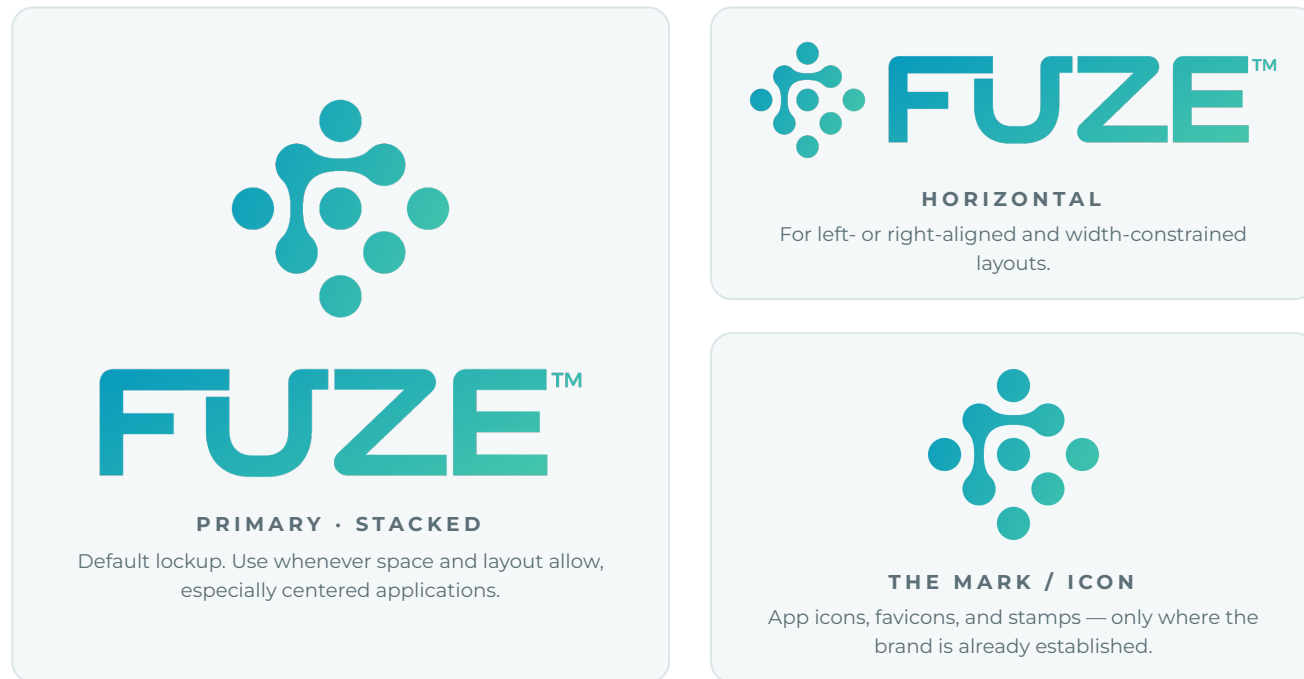
- **Confident, not boastful.** Let third-party validation carry the weight.
- **Precise, not hype.** Specific standards beat superlatives.
- **Clean & modern.** Short sentences. Active voice. No jargon walls.

## APPROVED ONE-LINE DESCRIPTION — COPY &amp; PASTE

“FUZE® is a non-leaching antimicrobial textile treatment that bonds proprietary metamaterial permanently into fibers during standard finishing — OEKO-TEX® Class I, bluesign® approved, EPA-registered, and PFAS-free.”



The FUZE® identity pairs the **molecular mark** — a bonded cluster that signals our metamaterial chemistry — with a custom rounded wordmark. Always reproduce the logo from approved master artwork. Never recreate, redraw, or re-typeset it.



## APPROVED COLOR TREATMENTS



Use the gradient lockup as the default. Use the white knockout on dark or photographic backgrounds; use the solid navy mark for single-color or low-cost reproduction. The mint single-color version is available for tonal applications on request.





## CLEAR SPACE & MINIMUM SIZE



Maintain clear space equal to the height of the mark's central cluster on all sides. Keep all other logos, type, and graphics outside this zone.



Horizontal — min **1.0"**  
wide



Icon — min **0.25"**

## CO-BRANDING — "TREATED WITH FUZE"

TREATED WITH



When FUZE® serves as the technology platform behind a partner brand, use the approved "Treated with FUZE®" lockup. The FUZE® mark should be **no smaller than 35%** of the partner's primary logo. Custom co-branded hangtags are available through the FUZE® Marketing Team.

## NEVER DO THIS

**Don't** recolor the logo outside the approved palette.

**Don't** stretch, rotate, skew, or add effects/shadows.

**Don't** recreate the wordmark in another typeface.

**Don't** place the gradient mark on busy or low-contrast backgrounds — use the white knockout.

**Don't** separate the mark from the wordmark to invent new lockups.

Any off-guideline use must be approved in advance by the U.S. FUZE® Marketing Team. Request master artwork and approvals at [brand@fuze47.com](mailto:brand@fuze47.com).



The FUZE® palette is anchored by three core colors and the signature cyan-to-mint gradient. Reproduce these values exactly. Do not produce materials in colors outside this palette without written consent from the FUZE® Marketing Team.

### Pantone 309 C

HEX #043D4C  
RGB 4 · 61 · 76  
CMYK 100 · 64 · 52 · 45

### Pantone 312 C

HEX #03A7CE  
RGB 3 · 167 · 206  
CMYK 96 · 6 · 15 · 0

### Pantone 3255 C

HEX #48C0BB  
RGB 72 · 192 · 187  
CMYK 64 · 0 · 32 · 0

#### SIGNATURE GRADIENT

312 C Cyan → 3255 C Mint · 120° linear · reserved for hero surfaces, covers, and the logo mark

#### USAGE HIERARCHY

- **Navy 309 C** — primary text, headlines, and dark surfaces.
- **Cyan 312 C** — primary accent: links, rules, key figures, CTAs.
- **Mint 3255 C** — secondary accent and gradient endpoint.
- **Gradient** — covers, section bands, and the logo only.

#### NEUTRALS & SUPPORT



Ink #22323A · Slate #5C6E76 · Line #DCE6E9 · Wash #F4F8F9.  
Use neutrals for body copy, captions, dividers, and panel fills to keep the brand colors high-impact.



**Montserrat** is the FUZE® typeface across all weights. Its geometric, rounded forms echo the wordmark. Use Montserrat for everything; where it is unavailable (e.g. email clients), substitute Arial or system sans-serif.

# Aa

## Montserrat

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z  
a b c d e f g h i j k l m n o p q r s t u v w x y z 1 2 3 4 5 6 7 8 9 0

Light 300   Regular 400   Medium 500   **SemiBold 600**   **Bold 700**   **ExtraBold 800**

### TYPE SCALE & ROLES

| ROLE             | SPECIFICATION                      | SAMPLE                                    |
|------------------|------------------------------------|-------------------------------------------|
| Display / H1     | ExtraBold 800 · 24–30pt            | <b>Brand &amp; Compliance</b>             |
| Section / H2     | ExtraBold 800 · 16–18pt            | <b>How We Talk About FUZE</b>             |
| Kicker / Eyebrow | Bold 700 · 8pt · +0.24em · UPPER   | <b>PART ONE · COMPLIANCE</b>              |
| Body             | Regular 400 / Medium 500 · 10–11pt | Clear, precise sentences carry the brand. |
| Caption / Legal  | Medium 500 · 7.5–8.5pt · Slate     | Third-party reports available on request. |



## HANGTAGS & PACKAGING



FUZE® provides a standard hangtag to explain the technology. Partners may also commission a custom co-branded hangtag through the Marketing Team. All hangtag copy must use approved (green) claims only.

**Approval required.** All packaging, hangtags, and on-product copy must be pre-approved before production.

## DIGITAL & LINKING

When using the FUZE® name or marks online — website, social, e-commerce, or press — include the following attribution:

*“For more information on FUZE®, please visit [fuze47.com](https://fuze47.com).”*

## OFFICIAL DESTINATIONS

| PROPERTY                | URL                                               |
|-------------------------|---------------------------------------------------|
| Corporate & brand       | <a href="https://fuze47.com">fuze47.com</a>       |
| Partner portal (Atlas)  | <a href="https://fuzeatlas.com">fuzeatlas.com</a> |
| Public FAQ & calculator | <a href="https://fuzefaq.com">fuzefaq.com</a>     |

**Retired:** the legacy domain [fuzebiotech.com](https://fuzebiotech.com) from the 2021 edition should no longer be referenced in new materials.



PART THREE · PRODUCT & PARTNERSHIP

# One Chemistry, Four Tiers — Permanent at Every Level

FUZE® is offered in four tiers set by application dose. **Every tier bonds permanently and does not wash out.** Tiers differ in antimicrobial test performance and the **ancillary benefit stack** — not durability. F1 is the flagship with the full benefit set; lower tiers are priced for value, not lower in quality.

| TIER               | DOSE       | ANCILLARY BENEFIT STACK                                                                                               | POSITIONING                      |
|--------------------|------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------|
| F1 · Full Spectrum | 1.0 mg/kg  | Full stack — antimicrobial + reduced water surface tension, faster drying, UV & color protection, microfiber benefits | Flagship · all benefits          |
| F2 · Advanced      | 0.75 mg/kg | Antimicrobial + selected ancillary benefits                                                                           | Strong performance, better price |
| F3 · Core          | 0.5 mg/kg  | Antimicrobial protection · core benefits                                                                              | Everyday value                   |
| F4 · Foundation    | 0.25 mg/kg | Antimicrobial protection · essential                                                                                  | Best price point                 |

**Durability is not a tier setting.** The metamaterial bonds permanently into the fiber at every application level; longevity is a function of the bond and the *substrate*, not the dose. Antimicrobial performance is third-party tested (AATCC 100 / ASTM E2149 / ISO 20743); reports available to partners on request.

### HOW FUZE WORKS

During finishing, FUZE® metamaterial bonds permanently into the fiber surface — **at every tier**. Protection is **contact-based and non-leaching** — the bonded material disrupts microorganisms on contact rather than releasing chemistry into the wash water. The bond does not wash out (durability follows the fiber/substrate, not the dose), with no effect on hand feel, color, breathability, or drape.

### THREE APPLICATION METHODS

- **Exhaust (dyebath)** — ideal for knits and piece-dyed fabric.
- **Pad-dry-cure** — for wovens and high-volume production.
- **Spray** — for nonwovens, composites, and localized treatment.

Standard finishing equipment · cure 150–170°C · water-based · no shelf-life limit.



FUZE® is **non-leaching by design** — which means the *right test* matters. Our lead method is **ASTM E2149**, the dynamic-contact test built for non-leaching antimicrobials. This is our clearest point of technical difference, and partners should state it with confidence.

**ASTM E2149 — OUR LEAD TEST**

Treated fabric is shaken in a bacterial suspension; reduction is measured after dynamic contact. It rewards direct surface contact — exactly how FUZE® works. No ion cloud required.

**AATCC 100 — HISTORICAL METHOD**

Stacks fabric layers around an inoculated coupon. It was designed for *leaching* chemistries, whose ions saturate the layers. FUZE® still passes at higher tiers (F1/F2), where density overcomes the test's geometry.

**THE POSITION — PARTNERS MAY USE VERBATIM**

*“We test on ASTM E2149 because it’s the method designed for non-leaching antimicrobials. Leaching chemistries rely on AATCC 100 because its geometry helps their ions saturate the layers. FUZE® doesn’t leach, by design. Meet us on the right test.”*

**THE RIGHT TEST BY TIER**

| TIER    | PRIMARY METHOD         | WHY                                                                            |
|---------|------------------------|--------------------------------------------------------------------------------|
| F4 / F3 | ASTM E2149             | Dynamic-contact test — direct mechanism match for non-leaching chemistry.      |
| F2 / F1 | ASTM E2149 + AATCC 100 | Sufficient metamaterial density to pass both, including the layered AATCC 100. |

**FULL STANDARDS BATTERY**

Antimicrobial efficacy: **ASTM E2149**, AATCC 100, ISO 20743 · Antifungal: **AATCC 30** · Antiviral: **ISO 18184** (data only) · Substrate & skin: OEKO-TEX® Class I, bluesign®.

**Say “third-party validated,” not “EPA-validated.”**  
Our advantage is that we *share* independent lab reports — competitors self-publish wash claims and don’t.



To maintain consistent messaging, performance, and quality across every partner, the following items require **FUZE® pre-approval** before treatment, application, or commercialization.

## REQUIRES PRE-APPROVAL

### Fabric & treatment

- Fabric submission & specifications
- Anticipated minimums & volumes
- Treatment tier (F1–F4) selection

### Marketing & brand

- Hangtags & on-product copy
- Web, social & e-commerce claims
- Domain linking & co-branded assets

### Partnership

- Environmental & philanthropic initiatives
- Press releases naming FUZE®

## HOW WE SUPPORT YOU

- **FUZE® Atlas partner portal** — submit fabrics, track tests, review approved results, book meetings.
- **Marketing kits & collateral** — approved copy, imagery, and claim guidance.
- **Third-party test reports** — AATCC 100 / E2149 / ISO data on request.
- **Technical & regulatory support** — SDS, certifications, and compliance Q&A.
- **Account servicing** — a dedicated team executing brand agreements and timelines.
- **Trade-show & co-marketing** — shared booth and partnership programs.

**Validation on demand.** Testing standards, requirements, and current data are available through the Atlas portal — integrated into every brand agreement and certification.



Sustainability isn't a claim we add on — it's built into how FUZE® is made and how it performs.

From recycled-electronics feedstock to solar-capable production and PFAS-free chemistry, FUZE® is engineered to lower impact at every stage — and to extend the useful life of the textiles it protects.

## 0

### PFAS & FOREVER CHEMICALS

Completely PFAS-free, aligned with tightening U.S. and EU restrictions.

## PPM

### ACTIVE INGREDIENT DOSE

Effective at parts-per-million; trace wastewater levels well below limits.

## ∞

### SHELF LIFE • NO EXPIRY

FUZE® does not expire — factories hold stock with zero spoilage waste.

#### HOW WE LOWER IMPACT

- **Recycled feedstock** — source material reclaimed from recycled electronics.
- **Solar-capable manufacturing** in Salt Lake City, Utah.
- **Drop-in process** — adds minimal energy to existing finishing; no new machinery.
- **Non-leaching** — designed to keep active material in the fabric, not the wash water.

#### LONGER LIFE, LESS WASTE

- **Extends garment life** by controlling odor and microbial deterioration — fewer textiles discarded.
- **Circularity-compatible** — does not impair recyclability of base textiles.
- **ESG-aligned** — supports corporate sustainability goals and the UN SDGs for health and responsible production.

Sustainability and lifecycle comparisons vs. specific competitor chemistries are available to partners on request.





## PART FOUR · GLOBAL COMPLIANCE &amp; MARKET CLAIMS

## One Principle, Eight Markets

Across every major market the same line holds: a FUZE®-treated textile may claim to protect **the article itself** — odor, freshness, resistance to mildew and deterioration — but never to protect **the wearer** from disease. What changes is the *mechanism* behind the active and the exact wording each regulator accepts.

## PRIMARY MARKETS — PAGES 18–21

| MARKET         | GOVERNING FRAMEWORK                                        | YOU MAY CLAIM                                                                        | KEY COMPLIANCE HOOK                                                                             |
|----------------|------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| United States  | EPA / FIFRA — Treated Article Exemption (PR Notice 2000-1) | Inhibits odor-causing bacteria; resists mildew/deterioration; freshness & durability | Active federally EPA-registered · no kill/public-health claims · no “safe/non-toxic”            |
| European Union | BPR (EU) 528/2012, Art. 58 — Treated Articles              | The article’s own biocidal property (odor / freshness / material protection)         | Active approved or under BPR review for PT9 + supplier on Article 95 list · Art. 58(3) labeling |
| Canada         | PCPA / PMRA — Treated Articles (2023 PCPR)                 | “Treated... to control odours”; resist bacterial odours; mildew/deterioration        | Preservative PMRA-registered · claims qualified · non-public-health disclaimer                  |
| China          | GB/T 20944 (test) + GB/T 31713 (safety)                    | An “antibacterial (抗菌)” grade tied to the measured GB test result                    | Test to GB/T 20944.3 (shake-flask) + meet GB/T 31713 · label the grade honestly                 |

## EXPANSION MARKETS — PAGES 22–25

| MARKET         | MODEL                                      | ROUTE TO CLAIM                                                                      |
|----------------|--------------------------------------------|-------------------------------------------------------------------------------------|
| United Kingdom | GB BPR (HSE) — treated articles            | Active approved/under GB review · NI follows EU BPR · <b>2026 deadline — verify</b> |
| Japan          | SEK Mark (JTETC) — voluntary certification | Certify via JIS L 1902 · display the SEK Mark category                              |
| Korea          | K-BPR (MOE) — treated articles             | Active approved/notified under K-BPR · test to KS K 0693                            |
| Australia      | APVMA (Agvet Code)                         | Article-protection claims generally exempt · health claims → APVMA/TGA              |

† The FUZE® active substance is identified by chemical name and CAS number in the product SDS. This guide is directional; confirm market-specific status with FUZE® and local regulatory counsel before publishing claims in a new market.



## THE FRAMEWORK

FUZE® is federally EPA-registered as an antimicrobial. A textile treated with it is an **exempt treated article** (40 CFR 152.25(a) / EPA PR Notice 2000-1) — provided every claim stays on protecting the article and the treatment follows the registered use.

### THE COMPLIANCE HOOK

The registration lives with the *active*, not the textile. The moment downstream copy claims to protect *people* (kill rates, pathogens, disease), the textile becomes an unregistered public-health pesticide — the exact FIFRA exposure we use against competitors.

**The “safe / non-toxic” trap.** FIFRA §25(j) prohibits “safe,” “non-toxic,” and “harmless” on anything with pesticidal registration. Use *OEKO-TEX® Class I* and *PFAS-free* instead of blanket safety claims.

### ● You may say

- Inhibits the growth of odor-causing bacteria
- Inhibits the growth of mildew/mold that deteriorates the article
- Keeps fabric fresher, longer · durable, non-leaching
- PFAS-free · OEKO-TEX® Class I · factual certifications

### ● You may not say

- Antibacterial / germicidal · “kills 99.9%”
- Named pathogens (MRSA, Staph, E. coli)
- Protects the wearer / disease / virus claims
- “EPA-approved/validated” durability · “safe” · “non-toxic”

**Source:** EPA PR Notice 2000-1; 40 CFR 152.25(a); FIFRA §25(j). See also the U.S. claims framework on pages 05–06.



## THE FRAMEWORK

Under the EU Biocidal Products Regulation (528/2012), a FUZE®-treated textile is a **“treated article” (Art. 58)**. It may be placed on the EU market provided the FUZE® active† is **approved — or still under assessment in the BPR Review Programme — for product-type 9** (fibre/textile preservatives), and the active’s supplier is on the **ECHA Article 95 list**.

### MANDATORY LABELING — ART. 58(3)

When any biocidal property is claimed, the article must be labeled with the claim, the **name of the active substance(s)**, and relevant instructions. Suppliers must answer a consumer’s information request **within 45 days**.

### ● You may say

- The article’s own biocidal property — odor & freshness control
- Resists microbial deterioration of the fabric
- Durable, non-leaching, PFAS-free
- ... with the required active-substance labeling

### ● You may not say

- Human-health protection · disease / virus claims
- Biocidal benefit beyond the article itself
- Any implication the textile is a registered biocidal product

## BEFORE SELLING IN THE EU — CONFIRM WITH COUNSEL

Verify (1) the FUZE® active’s current PT9 status in the BPR Review Programme, (2) Article 95 listing for the active/supplier, and (3) any classification & labeling outcome. EU active-substance reviews are live and can change market-access conditions.

**Source:** Regulation (EU) 528/2012, Art. 58 & 95; ECHA Treated Articles guidance. † Active identity in SDS.



## THE FRAMEWORK

Under Canada's *Pest Control Products Act*, a FUZE®-treated textile is an **exempt treated article** when: (1) the FUZE® preservative is registered/authorized in Canada, (2) use is limited to protecting the article, and (3) it is applied per the registered rates and methods. Claims must match the preservative's approved label and be **qualified** — “antimicrobial” or “preservative” may never stand alone.

**Required where confusion is possible:** “This product does not protect users or others against bacteria, viruses, germs or other disease-causing organisms.”

### ● PMRA-accepted textile claims

- “Treated with an antimicrobial agent to **control odours.**”
- “Treated to **resist the development of bacterial odours.**”
- “Extends the useful life of the article by controlling **deterioration caused by mould/mildew.**”
- “Contains a preservative to **mitigate the development of odours.**”

### ● You may not say

- “Antimicrobial” / “preservative” unqualified
- Pathogens not on the active's registered label
- “Prevents infection” or any health benefit

**Source:** PMRA, *Acceptable Claims for Articles Treated with Antimicrobial Preservatives* (12 Mar 2024, §3.4); PCPR amendments (in force 5 Jun 2023); DIR2016-01. Claims above are verbatim PMRA examples and are conditional on the FUZE® preservative being registered in Canada for this use.



China uses a **performance-grade** model rather than pesticide registration: a textile is tested to a national **GB** method, and an “**antibacterial (抗菌)**” grade may then be labeled honestly against the measured result — provided the product also meets the GB safety standard.

#### THE GOVERNING STANDARDS

| STANDARD     | METHOD                                                             | NOTE                                                                                              |
|--------------|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| GB/T 20944.1 | Agar diffusion (qualitative)                                       | Screening for antibacterial activity.                                                             |
| GB/T 20944.2 | Absorption (quantitative)                                          | Static quantitative method.                                                                       |
| GB/T 20944.3 | Shake-flask / oscillation (quantitative — <b>dynamic contact</b> ) | <b>FUZE's strongest method</b> — rewards non-leaching contact-kill. China's analog to ASTM E2149. |
| GB/T 31713   | Hygienic safety requirements                                       | Safety layer for antibacterial textiles.                                                          |

#### WHY THIS FAVORS FUZE

The shake-flask method (GB/T 20944.3) is a **dynamic-contact** test — the same mechanism logic as ASTM E2149. It is the method that best demonstrates FUZE®'s non-leaching, contact-based performance. Lead with it in China the way we lead with E2149 elsewhere.

#### CLAIMING IN CHINA

- State the **antibacterial grade** achieved under GB/T 20944, no more.
- Meet **GB/T 31713** safety requirements before labeling.
- Antiviral data via **ISO 18184** — technical reference only.
- No disease-protection or kill-rate marketing claims.

**Source:** GB/T 20944.1/2/3; GB/T 31713-2015. **To confirm:** the specific updated GB standard version with the FUZE® lab partner before citing it as binding.



## THE FRAMEWORK

Post-Brexit, Great Britain runs its own **GB Biocidal Products Regulation** (administered by the **HSE**), which mirrors the EU treated-article rules: a FUZE®-treated textile may be placed on the GB market only if the FUZE® active† is approved or under assessment in the **GB Review Programme** for the relevant product-type, with the same Art. 58-style labeling and claim limits.

### TWO REGIMES, ONE ISLAND GROUP

**Great Britain** → GB BPR (HSE). **Northern Ireland** → **EU BPR** under the Windsor Framework. Plan GB and NI separately — an EU-compliant route covers NI but not GB.

### ● You may say

- The article's own biocidal property — odor & freshness
- Resists microbial deterioration of the fabric
- Durable, non-leaching, PFAS-free · with required labeling

### ● You may not say

- Human-health protection · disease / virus claims
- Biocidal benefit beyond the article itself

**Live deadline — confirm before GB entry.** The HSE has begun issuing **non-approval** decisions under the GB Review Programme; affected antimicrobial-treated articles must stop being placed on the GB market by **28 September 2026**. Confirm the FUZE® active's current GB status with UK regulatory counsel before committing to the GB market.

**Source:** HSE — GB Biocidal Products Regulation & Northern Ireland overview; GB BPR review decisions (2026). † Active identity in SDS.



Japan uses a **voluntary performance-certification** model. The recognized credential for antibacterial, deodorant, and antiviral textiles is the **SEK Mark**, issued by JTETC (Sengikyo) and accredited to ISO/IEC 17065. It is earned through **JIS L 1902** testing — not pesticide registration.

#### SEK MARK CATEGORIES & METHODS

| SEK CATEGORY              | TEST METHOD            | CLAIM BASIS                                                           |
|---------------------------|------------------------|-----------------------------------------------------------------------|
| Antibacterial & Deodorant | JIS L 1902             | Controls bacterial growth & resulting odor on the fabric.             |
| Antimicrobial (制菌)        | JIS L 1902             | Higher-bar bacterial control on the textile.                          |
| Antiviral                 | JIS L 1922 / ISO 18184 | Reduces specific viruses on the fabric (data-based, no health claim). |

#### THE ROUTE TO MARKET

Test the FUZE®-treated fabric to JIS L 1902 through a JTETC-affiliated lab, certify the appropriate SEK category, then display the SEK Mark. The mark itself carries the credibility — Japanese buyers look for it.

#### CLAIMING IN JAPAN

- Use the **SEK Mark** and its category wording — nothing beyond it.
- Antiviral = fabric-surface data only; never disease protection.
- Keep performance figures in technical references.

**Source:** JTETC (Japan Textile Evaluation Technology Council) — SEK Mark certification standards; JIS L 1902.



## THE FRAMEWORK

South Korea's **Consumer Chemical Products and Biocides Safety Act (K-BPR)**, administered by the Ministry of Environment, is modeled on the EU BPR and explicitly covers **biocide-treated articles**. A FUZE®-treated textile may be sold only if the FUZE® active† is **approved/notified under K-BPR** for the relevant product-type, with K-BPR labeling and advertising controls.

### PERFORMANCE TESTING

Antibacterial performance is demonstrated to **KS K 0693** (Korea's antibacterial textile test, analogous to JIS L 1902).

### You may say

- Article's own antibacterial / deodorizing property
- Freshness & material-protection benefits
- ... with the K-BPR-required treated-article labeling

### You may not say

- Human-health / disease protection
- Claims using a non-approved biocidal active
- Anything beyond the approved K-BPR scope

**Confirm before Korea entry.** K-BPR is a registration-style regime — the FUZE® active must be approved or notified for its product-type before any treated-textile is placed on the Korean market. Confirm status with Korean regulatory counsel.

**Source:** Consumer Chemical Products and Biocides Safety Act (K-BPR), Ministry of Environment (in force 2019); KS K 0693. † Active identity in SDS.





### THE FRAMEWORK

Australia regulates antimicrobial chemistries under the **Agvet Code** (APVMA). The dividing line is the claim: a textile treated to **protect itself** — odor control and material preservation (“bacteriostatic”) — generally does not require product registration, while claims to **protect people** from microbes or disease move the product into APVMA registration and/or the **TGA** (therapeutic goods).

#### THE SAME PRINCIPLE, AGAIN

Protect the *article* → freely claimable. Protect the *person* → regulated. Keep FUZE® messaging firmly on the article side.

#### ● You may say

- Controls odor-causing bacteria on the fabric
- Resists deterioration & staining from mould/mildew
- Keeps the textile fresher, longer · PFAS-free

#### ● You may not say

- Protects the wearer / hygiene / sanitiser claims
- Kills germs, viruses, or named pathogens
- Any therapeutic / disease-prevention benefit

**Confirm:** Australia's treated-article position is less codified than the U.S./EU. Confirm the classification of FUZE®-treated textiles and any registration trigger with the APVMA (and TGA, if a health benefit is ever implied) before entering the market. **Source:** APVMA — Agvet Code / chemicals we regulate.



## THE PATH TO PRODUCTION

### 1 Consult

Identify fabrics, target tier, and performance goals with your FUZE® rep.

### 2 Sample

Treat sample fabric at a partner factory (typically 2–4 weeks).

### 3 Validate

Independent lab testing of treated samples (1–2 weeks).

### 4 Approve

Review results and lock production specifications & brand assets.

### 5 Produce

Begin production-scale treatment with ongoing QC verification.

## SUBMIT & GET APPROVALS

Start a brand application, request marks, or submit copy for claim approval through the partner portal or your account team.

- Brand application & fabric submission
- Logo / master artwork requests
- Claim & collateral pre-approval
- Test reports & certifications



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Questions about whether specific language is approved? Always ask before you publish — [brand@fuze47.com](mailto:brand@fuze47.com). This guide supersedes the 2021 Brand Compliance edition (Version 1.0).



BUILDING A CLEANER,  
HEALTHIER WORLD — ONE TEXTILE AT A TIME